

PRODUCT BRIEF · v1.2 · JUNE 2026

AI Ticket Auto-Resolution

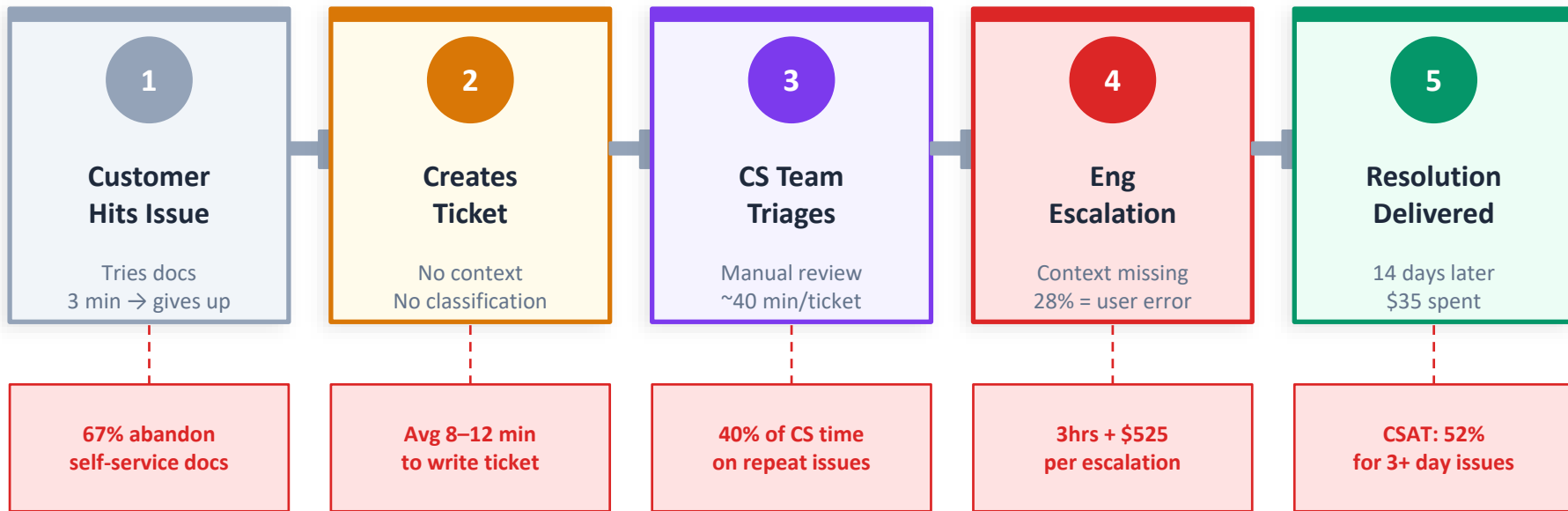
Implementation Update

What we built · What's live · What's next



Mile · Ticket Auto-Resolution

How Every Ticket Gets Resolved Today



Total journey cost: \$22–35 per ticket · \$175/hr engineering time · 2-week SLA exceeded for every issue regardless of complexity

Three Users, One Broken System

PRIMARY USER

Maya

Operations Manager

2.3

tickets / month

"I just need this fixed. I don't have time to read a 40-page doc."

- Docs don't match her error
- 14-day resolution wait
- No status visibility after submit

CS TEAM

Alex

Customer Success Manager

40%

time on repeat issues

"Half my queue is the same 5 issues. I could fix those in my sleep."

- Every ticket starts from scratch
- No AI-assist on obvious fixes
- Manual routing wastes 18% of time

ENG TEAM

Dev Team

Engineering & Product

28%

escalations = user error

"We keep getting pulled in for things that aren't actually bugs."

- Sprint disruption from support
- Tickets arrive with no context
- \$175/hr cost for user errors

The Current State Is Broken

Industry benchmarks — Forrester, HDI, Zendesk CX Trends 2024–25

14 Days

Avg. time to resolution

SLA across all ticket types

67%

Self-service abandon rate

Give up on docs within 3 min

\$35

Cost per manual ticket

Fully loaded CS + Eng (HDI)

Root Causes

- 35% of all tickets are repeat known issues that could be auto-resolved
- 28% of Eng escalations are user errors — wrong setting, missed step, misunderstood feature
- 40% of CS time spent on problems that have an existing documented fix

Not All Tickets Are Equal

Three resolution paths — each matched to issue complexity

TYPE 3

~45% of tickets

Solvable Now

User just needs a nudge — wrong setting, missed step, misunderstood feature.

e.g. Booking widget missing after publish

Resolution →

AI self-serves in chat → 0 human touch → auto-closes after confirm

Target SLA: < 15 min · Cost ~\$0

TYPE 2

~35% of tickets

Known Issue

Already in backlog. The fix takes time, but a workaround exists right now.

e.g. Analytics dashboard blank on Chrome 124+

Resolution →

AI acknowledges + delivers workaround + ETA from KB → CS reviews

Target SLA: < 2 hrs · Cost ~\$5

TYPE 1

~20% of tickets

Novel Issue

First occurrence. Requires investigation. Could be a new bug or edge case.

e.g. Salesforce lead sync stopped after 3.13

Resolution →

AI classifies + proposes fix → CS approves → Eng escalates with full context

Target SLA: < 8 hrs · Cost ~\$15

What's New — 6 Capabilities Added

The demo runs end-to-end on bundled data · numbers are illustrative until measured

Free-text Classifier

BEFORE: 3 scripted scenarios → AFTER: Any natural language input

 Live and wired in the demo

KB → Chat Wired

BEFORE: Disconnected → AFTER: Add article → AI uses it instantly

 Live and wired in the demo

Feedback → Dashboard

BEFORE: Static numbers → AFTER: Live counters update on every action

 Live and wired in the demo

Notifications Center

BEFORE: None → AFTER: Bell + badge + 'notify me' flow

 Live and wired in the demo

Media + Agent Assist

BEFORE: Text only → AFTER: Attach image/video + human handoff

 Live and wired in the demo

Golden Set Tests

BEFORE: None → AFTER: 8 expert cases + custom prompts

 Live and wired in the demo

Customer Chat Interface

Customer describes issue in plain language — no forms, no categories

M

Ticket Auto-Resolution

AI

Customer

CS console

README

Architecture

✦

• Mile Assistant

• resolves most issues without a form

Try a scenario:

• Booking widget missing

• Analytics charts blank

• S

My booking widget disappeared from my hotel homepage.

ISSUE CLASSIFIED

• Type 3 · Solvable now

Booking engine · 94% confidence · Auto-resolve

Your booking widget is set to 'Draft' on the homepage layout.

Steps: 1. Open Content → Homepage → Layout 2. Find Booking widget block 3. Toggle visibility to Published 4. Click Publish

👍 This Fixed It

🗨 Still Broken

TICKET STATUS

Ticket

TCK-2041

Status

• In progress

Type

Type 3

Confidence

94%

Routing

Auto-resolve

A ticket never closes until you confirm the fix worked.

Free-text input — any issue, any phrasing

Scenario chips for guided demos

AI responds in seconds, not days

Ticket status panel — always visible, no black hole

Approval Queue

Every ticket sorted by confidence · routing updates live as thresholds change

M Ticket Auto-Resolution ♦ AI

Customer

CS console

README

Architecture

Approval queue

Knowledge base

Analytics

Golden set

Approval queue

Sorted by confidence · 6 awaiting review

 **2**
Auto-resolve
No human touch

 **2**
CS approval
CS approves or edits

 **1**
CS rewrite
CS full review

 **1**
Eng queue
Eng manual resolution

CONF.	TICKET	TYPE	PRI.	ROUTING	
96	Booking widget not visible after publish TCK-2041 · Maya Okonkwo	Type 3	P3	Auto-resolve	>
92	Cannot invite teammate — button greyed out TCK-2050 · Lena Fischer	Type 3	P3	Auto-resolve	>
88	Dashboard charts render blank in Chrome 124+ TCK-2044 · David Reyes	Type 2	P2	CS approval	>
79	Scheduled campaign sent twice to segment TCK-2046 · Priya Anand	Type 2	P3	CS approval	>
38	Salesforce lead sync stopped after 3.13 TCK-2048 · Jordan Blake	Type 1	P1	Eng queue	>

-  Auto-resolve: no human touch needed
-  CS approval: review AI draft, one click
-  CS rewrite: AI draft needs full edit
-  Eng queue: novel issue, full escalation

CS Approval — Ticket Detail & Knowledge Base Manager

CS sees the AI draft + match evidence + KB wired live into the classifier

← Back to queue

Type 3 Solvable now P3

Booking widget not visible after publish

Maya Okonkwo · Westmoreland Hotel · 2m ago · Booking engine

CONVERSATION

Maya Okonkwo: "My booking widget disappeared from my homepage after I published a new hero."
Mile Assistant: Asked a clarifying question, then classified as Type 3 (Solvable now) in Booking engine at 96% confidence.

AI-DRAFTED RESPONSE

Edit

Your booking widget reverted to 'Draft' visibility when you published the new hero section. Open Content → Homepage → Layout, set the Booking widget block to 'Published', then click Publish.

MATCH EVIDENCE

KB-118 · Widget visibility resets on layout publish 98%
37 past tickets resolved with this fix 94%

96%
AUTO

✓ Approve & send

Edit

↑ Escalate

M Ticket Auto-Resolution · CS console

Knowledge base

+ Add article

Articles that train confidence scores. More thorough articles → higher auto-resolve rates.

4 articles 148 AI uses this month

Booking widget disappears after hero publish

47 AI uses

booking engine widgets publish

Edit

Analytics dashboard blank on Chrome 124+

37 AI uses

analytics chrome charts

Edit

Email campaign sends twice on rapid segment edit

12 AI uses

email campaigns segments

Edit

Invite button greyed out at seat limit

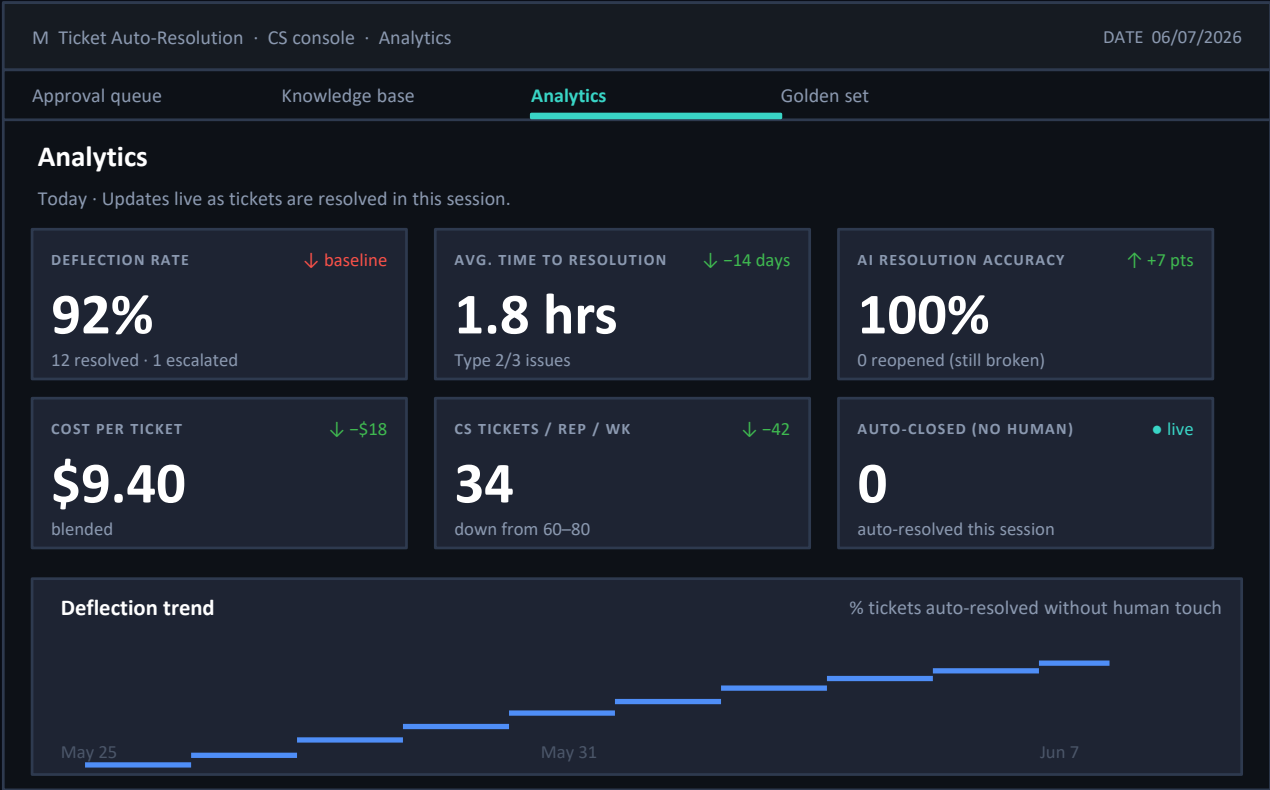
52 AI uses

account billing invites

Edit

Analytics Dashboard

Updates live as tickets are resolved in this session · seeded baseline + session activity



- 92% deflection — computed from session data + seeded baseline
- 1.8 hrs avg resolution (down from 14 days)
- \$9.40 / ticket blended cost (target <\$8)
- Dashboard numbers are illustrative — not yet externally measurable

Quality Gate & System Architecture

Regression suite runs live · architecture is the real system, not a diagram

REGRESSION SUITE

Golden Set Panel

100%
PASS RATE

8/8
PASSED

Expert-labeled scenarios running live through the classifier.

 All 8 test cases are passing. The classifier is fully aligned.

ID	TEST CASE	EXPECTED	STATUS
G-01	Booking widget missing after publish	Type 3 96%	✓ Pass
	TYPE 3 SOLVABLE Matches KB-001		
G-02	Analytics dashboard blank in Chrome 124+	Type 2 96%	✓ Pass
	TYPE 2 KNOWN Matches KB-002		
G-03	Email campaign sent duplicate to segment	Type 2 96%	✓ Pass
	TYPE 2 KNOWN Matches KB-003		
G-04	Invite button greyed out at seat limit	Type 3 96%	✓ Pass
	TYPE 3 SOLVABLE Matches KB-004		
G-05	Salesforce sync stopped after 3.13	Type 1 38%	✓ Pass
	TYPE 1 NOVEL Matches KB-005		

ARCHITECTURE

System Architecture

Three-Layer Architecture

Presentation layer

- TicketResolutionComponentshell, view switcher, tabs, notifications
- CustomerChatComponentguided customer resolution flow



State + logic layer

- DemoStateService
single shared live state
- routeFor(score, thresholds)
confidence to route mapping



Data layer

- ticket-data.ts
scenarios, queue, KB, metrics
- expert-labeled test cases

.NET 8 Backend (built, not deployed) — IssueClassifier interface ready for ML / LLM+RAG swap

The Honest Scorecard

What's real, what's seeded, what must still be built before production

✓ REAL TODAY

Built and wired end-to-end

8

capabilities live

"Taxonomy, routing, KB-to-chat, feedback loop, golden set — all working in the demo."

- Free-text classifier (transparent rule-based)
- KB Manager → resolver in real time
- Feedback loop → live dashboard counters
- Notifications bell + 'notify me' flow
- Media attach + agent assist handoff
- Golden set regression suite (8/8 passing)

~ ILLUSTRATIVE

Looks real — not yet measured

63%

dashboard headline

"Numbers are computed from session activity + seeded baseline. Do not quote externally."

- Deflection rate is session-driven
- AI accuracy is a moving target
- Cost / CSAT are placeholders
- Resolution times are seeded
- All metrics need real measurement study

✗ NOT BUILT YET

Production gates — must do

0

real measurements yet

"Before quoting accuracy externally, replace the heuristic with grounded ML / LLM+RAG."

- Grounded ML / LLM+RAG classifier
- Baseline measurement study
- Ticketing system-of-record integration
- PII / data-residency policy
- Role-gated threshold sliders (admin only)

Human Approval at Every Gate

No ticket closes without the right sign-off. AI amplifies CS — it does not replace them.

Confidence → Routing

CONFIDENCE ACTION		HUMAN ROLE	AUTO-CLOSE?
≥ 90%	Auto-resolve + send	None (Type 3 only)	Yes — after confirm
75–89%	AI drafts, CS approves	CS: approve or edit	Yes — after confirm
50–74%	AI drafts, CS rewrites	CS: full review & rewrite	No — CS manually closes
< 50%	No suggestion shown	Eng queue: full manual	No — Eng manually closes

4 Human Approval Gates

- 1

CS Approval Gate
CS reviews AI draft (75–89%) before customer sees it. Target: <30 min.
- 2

Customer Confirms
Ticket does NOT close until customer confirms fix worked. 'Still broken' reopens.
- 3

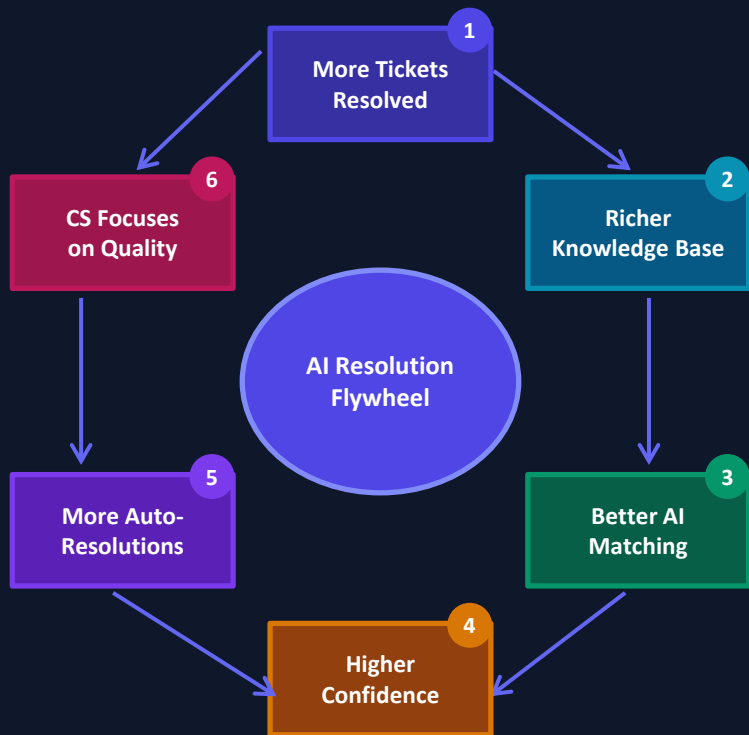
Eng Review
Novel (Type 1) or <50% confidence auto-routes to Eng with full context.
- 4

Escalation SLA
Escalated tickets require Eng acknowledgment within 4 business hours.

The Resolution Flywheel



A Self-Reinforcing Loop



1. More Tickets Resolved

Every closed ticket is a labeled training point — CS edits, customer confirmations, re-opens all captured.

2. Richer Knowledge Base

Resolutions populate the KB in real time. Known issues get workarounds. New patterns get flagged.

3. Better AI Matching

Denser KB = more accurate classification. Type 3 caught earlier; Type 1 matched to past tickets faster.

4. Higher Confidence

Better matching → AI is more certain. More tickets cross the auto-resolve threshold autonomously.

5. More Auto-Resolutions

Deflection rises. CS bandwidth frees up. SLA compresses from 14 days → hours for Type 2/3.

6. CS Focuses on Quality

Freed from triage, CS invests in KB curation & refinement — spinning the wheel even faster.

Success Metrics: How We Know It's Working

OKRs — measured from baseline, not from seeded demo numbers

Deflection Rate

tickets closed w/o Eng

NOW

~5%



TARGET

≥65%

Avg. Resolution Time

Type 2/3 issues

NOW

14 days



TARGET

< 2 hrs

Customer CSAT

on support interactions

NOW

52%



TARGET

≥78%

Cost per Ticket

fully loaded CS + Eng

NOW

\$35



TARGET

< \$8

CS Tickets / Rep / Wk

bandwidth freed

NOW

60–80



TARGET

< 35

AI Accuracy Rate

no re-open in 7 days

NOW

N/A



TARGET

≥90%

What We Build, When

0

Wks 1–4

Phase 0: Foundation

- Ownership matrix
- KB schema design
- Confidence model
- CS team training

Gate:

KB has ≥ 100 entries covering top issues

1

Months 1–3

Phase 1: MVP

- NL intake
- Issue classifier
- Type 3 auto-resolve
- Type 2 KB match

Gate:

Deflection rate $\geq 25\%$ after 30 days

2

Months 4–6

Phase 2: Approval Queue

- CS approval UI
- Type 1 draft proposals
- Priority routing
- Eng escalation context

Gate:

CS approval SLA < 30 min; accuracy $\geq 80\%$

3

Months 7–9

Phase 3: Learning Loop

- Feedback signals
- KB auto-update
- Analytics dashboard
- Ownership automation

Gate:

Deflection $\geq 55\%$; cost/ticket $< \$12$

Key Decisions Before We Build

PHASE 1 — Resolve Before

Q1

Who owns the AI knowledge base operationally — CS Lead, dedicated KB manager, or rotating responsibility?

Q2

What is the data residency and PII policy for storing customer conversations used in AI training?

Q3

Which ticketing system (Jira, Zendesk, Linear) is the system of record — and what does the integration look like?

PHASE 2 — Resolve Before

Q1

What is the escalation SLA for Type 1 Eng tickets, and who owns SLA enforcement?

Q2

Does the customer experience live inside the product (in-app chat) or as a separate support portal?

Q3

How do we handle P1 incidents where multiple customers hit the same issue simultaneously?

PHASE 3 — Longer-Term

Q1

At what deflection rate does it make sense to redeploy CS toward proactive customer success?

Q2

Should we publish real-time issue status publicly (like a status page) or keep it per-ticket?

Q3

How does this system interact with churn prediction and product-led growth signals?

Must Fix Before Public Launch

The demo proves the experience. These four close the gap to a defensible production launch.

1

Replace the rule-based classifier

Swap in a grounded ML / LLM+RAG model that cites sources for every auto-resolution. The `IssueClassifier` interface is already built — this is a drop-in.

Eng Lead

2

Run a baseline measurement study

Replace seeded dashboard numbers with measured values before any external claim. This is the only path to claiming '65% deflection' externally.

PM + Eng

3

Grow the Golden Set to 100+ cases

CS-expert sign-off on each case. Gate every release on pass-rate not regressing below the previous build.

CS Lead + PM

4

Wire to real ticketing system + lock PII policy

Jira / Zendesk / Linear integration. Lock down data-residency policy before customer conversations are stored. Role-gate the live threshold sliders.

Eng + Legal

THE ASK

The demo is live. Let's ship.

AI resolves 65%+ of tickets in under 2 hours — with zero loss of human control.

1

Align on open questions

KB ownership · data residency / PII policy · ticketing system of record

2

Phase 0 sprint

Build KB schema + ownership matrix + seed 100 entries

3

Baseline measurement study

Replace seeded metrics with real numbers before any external claim

4

Phase 1 kickoff

Schedule with CS Lead, PM, and Eng Lead — Classifier interface is ready

How We Know It's Working

Deflection Rate

tickets closed w/o Eng

NOW

TARGET

~5% -> ≥65%

Avg. Resolution Time

Type 2/3 issues

NOW

TARGET

14 days -> < 2 hrs

Customer CSAT

on support interactions

NOW

TARGET

52% -> ≥78%

Cost per Ticket

fully loaded CS + Eng

NOW

TARGET

\$35 -> < \$8

CS Tickets / Rep / Wk

bandwidth freed

NOW

TARGET

60-80 -> < 35

AI Accuracy Rate

no re-open in 7 days

NOW

TARGET

N/A -> ≥90%